

Preferences Become Measurable at Supervised Fine-Tuning

Phase 3 Results: Staged Attribution of Risk and Framing Behavior Across Post-Training

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1 Introduction

This project treats large language models as economic agents and recovers their preference primitives from their choices. Phase 1 fit a structural discrete-choice model to 1,200 controlled lottery choices from `claude-haiku-4-5` and found that the model prices the probability of winning and nearly ignores the size of the prize, with an extreme S-shaped Prelec weighting function ($\gamma = 2.69$). Phase 2 held terminal wealth fixed and varied only the wording, and found a violation of frame invariance: 40% versus 10% gamble-taking on arithmetically identical lotteries, with a stated-reference loss-aversion estimate of $\lambda = 3.234$ under $\phi = 1$. Both phases measured a single closed frontier model, and both ended on the same open question. Where do these properties come from?

Phase 3 takes it up by moving to open-weight models whose training stages are public. The identical Phase 1 and Phase 2 instruments are carried to matched base and post-trained checkpoints from three families, and choice probabilities are read directly from next-token logits rather than sampled. The decisive asset is OLMo-2-7B, which ships four public checkpoints along one training path: base, supervised fine-tuned (SFT), preference-optimized (DPO), and final Instruct. Running the same instruments across that staircase, in a single elicitation format, turns a coarse base-versus-instruct contrast into a stage-level attribution.

The headline is that supervised fine-tuning is the largest single step along the OLMo-2 training path, and the only step at which the structural preference estimates move. From base to final Instruct, dominance compliance rises from 0.555 to 0.896 and the frame gap from +0.099 to +0.349. SFT accounts for about three fifths of each change (dominance +0.200, 95% CI [+0.177, +0.226]; frame gap +0.144, CI [+0.133, +0.155]), and it carries the structural fit off the parameter bounds where the base model sits. After SFT the estimated parameters hold still: from SFT to DPO, $\Delta\lambda = -0.06$ with profile-likelihood CI [-0.23, +0.11] and $\Delta\gamma = +0.04$ with CI [-0.24, +0.34]. The later stages are still active. DPO adds +0.126 to dominance compliance and +0.075 to the frame gap, both clearly different from zero, and the final polish adds a small further amount. That later movement appears as a deteriorating structural fit (deviance on the same 1,600 cells rises from 229 at SFT to 457 at Instruct) rather than as shifted parameters, so preference optimization changes something the specification does not capture.

Where the profile comes from is a separate question, and the staircase answers it only in part. The OLMo base model places 0.555 of its probability on the larger of two sure amounts, close to a

coin flip, so SFT is the first stage at which the profile can be measured at all. These data cannot distinguish a profile that SFT installs from one planted in pretraining that SFT makes visible, which is the account Itzhak, Belinkov, and Stanovsky (2025) give for cognitive biases generally. What the staircase does establish is order. The measurable economic profile, and the structural parameters that summarize it, are in place before preference optimization begins. Two further results hold across families. Base models barely track value at all, consistent with the poor utility fits Wang, Li, and Chen (2025) report for the same Llama and Qwen pairs, and the dominance control shows why those fits fail. And the framing response is family-specific: Haiku suppresses gambling under loss wording, Qwen amplifies it sharply, OLMo amplifies it moderately, and Llama barely reacts.

Everything above comes from an immediate one-letter answer, and a reason-then-answer arm, run after the reviewer’s induced-valuation result, shows how much that restriction matters. Allowed to write out their reasoning before answering, the models follow an instructed expected-value rule on 0.965 of clear-cut cells (Qwen Instruct) and 0.803 (OLMo Instruct), against roughly 0.55 when answering immediately, and uninstructed they compute expected value on their own in most traces. The framing response changes with the regime. Qwen’s strong amplification under loss wording (+0.626) reverses to a small suppression (−0.135, 95% CI [−0.221, −0.053]), produced by one recurring accounting error: the model compares the gamble’s expected change in wealth with the sure option’s total. The staircase changes too. Under reasoning, OLMo SFT keeps a large frame gap (+0.375) that is nearly gone at the final Instruct checkpoint (+0.055), so the later training stages, invisible in the immediate-answer parameters, move behavior when the model deliberates. The attribution to supervised fine-tuning is a result about the immediate-answer regime.

2 The Result in Plain Language

An AI model is built in stages. First it is trained on a very large amount of text, producing a base model: a system that continues text plausibly and has never been taught to answer questions. Then it is fine-tuned, in several distinct steps, to behave like a helpful assistant. Most published work on AI decision-making tests only the finished assistant, which cannot tell you whether a quirk came from reading the internet or from the training that came after. One model family, OLMo-2, publishes a snapshot after each training step. So we asked the money questions from our earlier phases to the same model at four points in its life, in identical wording each time, and watched what changed.

The biggest change came at the first step. The base model was close to incoherent as an economic decision-maker. Offered a sure \$50 against a sure \$70, no gambling involved, it picked the larger amount barely more often than a coin flip. The base model does not have odd preferences; it has no preferences in any usable sense, and is producing a plausible-looking letter rather than comparing two amounts of money. After supervised fine-tuning, the first and simplest of the assistant-training steps, it picked the larger amount three quarters of the time and began reacting to how the choices were worded, and our model of its preferences settled on values that did not change afterward. The two later steps, the ones built on human preference comparisons, kept pushing in the same direction: the model became more reliable on the easy questions and somewhat more sensitive to wording, without changing the preferences our model recovers. The later stages are the ones people usually blame, since human feedback enters there. Here the measurable preferences are already in place before those stages begin. Whether supervised fine-tuning created them, or only made visible something the model absorbed from its reading, is a question the base model’s near-random

answers leave open.

Three other things came out of the same runs. The wording effect from Phase 2 has no single direction across systems. Claude Haiku became much more cautious when the downside was called a loss. Qwen became dramatically more willing to gamble under the same wording change. OLMo also became more willing, less dramatically. Llama hardly noticed. Four systems, four responses to one manipulation. The effect is a property of how a particular model was trained rather than of AI systems in general, so anyone reporting a framing bias from one model and calling it a fact about language models is reporting a fact about that model.

The answer also depends on how you ask, in a way that is model-specific and easy to miss. We put the same questions to each finished assistant in two standard formats. Qwen behaved sensibly in the chat format and collapsed into near-total refusal to gamble in the other. OLMo did the reverse. Llama was roughly stable in both. A study comparing two models in one fixed format may be comparing an engaged model against one that has checked out. The check is cheap: before analyzing anything, confirm the model reliably picks \$70 over \$50. If it does not, the rest of its answers are not preferences.

The third came from our reviewer, and it is the most uncomfortable. Experimental economists have a standard move: rather than guess what a subject wants, tell the subject what to want and pay them for it. We ran the instruction-only version, telling the models in plain language to be expected-value maximizers, to multiply payoff by probability and take the larger number. They could not do it. Compliance with the stated rule never rose above roughly 55% at any training stage, close to a coin flip, while the same models followed the part of the instruction requiring no arithmetic almost perfectly, reaching 100% on the easy questions. Asked for a harder rule involving a square root, one model’s compliance fell to 12% with a legible error pattern: it compared the transformed payoffs and dropped the probability entirely. So we let them work the problem out in writing before answering. Qwen then followed the rule 96% of the time and OLMo 80%. The obstacle was the one-letter format. With room to reason, the models’ reaction to loss wording changed as well, and Qwen’s reversed. Working through the arithmetic, it made the same bookkeeping mistake again and again, counting the gamble as a change in its money and the sure option as a total, and that mistake always favors the safe choice.

3 Experimental Design

3.1 Instruments, grid, and subjects

Phase 3 reuses the Phase 1 and Phase 2 instruments without modification. The exported grid (`data/phase3_grid.csv`) holds 1,680 design cells: 400 Phase-1 risk cells (40 procedurally generated contamination-safe gambles $\times$ 5 paraphrase templates $\times$ 2 counterbalanced orders), 1,200 Phase-2 framing cells (the same 40 gambles $\times$ 3 frames $\times$ 5 templates $\times$ 2 orders), and 80 dominant controls pitting a sure amount against a strictly larger sure amount. Every model sees byte-identical prompts, so nothing in the cross-model comparison is a difference in the instrument.

Three families, seventeen runs, 28,560 elicited choice probabilities. Qwen2.5-7B (base, Instruct under both formats); Llama-3.1-8B (base, Instruct under both formats), run once Hugging Face access to the gated repository was approved; and OLMo-2-1124-7B in four stages (base, SFT, DPO, Instruct) plus Instruct under the chat format. Six further induced runs cover the OLMo staircase under risk-neutral induction and Qwen Instruct chat under both induced rules. Six reason-then-answer runs (Section 3.4) cover Qwen Instruct chat and OLMo SFT and Instruct few-shot, each

uninduced and induced.

3.2 Elicitation by deterministic logits

Base models do not follow instructions, so the Phase 1–2 forced-tool-choice harness cannot run on them. Phase 3 reads the choice probability directly. Each prompt is wrapped in a $k = 4$ few-shot completion (exemplar answers balanced two A and two B, so no letter prior is induced) or passed through the model’s own chat template, and the full next-token distribution is read at the answer position:

$$P(\text{gamble}) = \frac{\sum_{t \in \mathcal{T}_g} P(t)}{\sum_{t \in \mathcal{T}_A} P(t) + \sum_{t \in \mathcal{T}_B} P(t)},$$

where $\mathcal{T}_A$ and $\mathcal{T}_B$ are the candidate answer tokens resolved per tokenizer, with and without a leading space. The unnormalized A/B mass $\sum_{\mathcal{T}_A} P + \sum_{\mathcal{T}_B} P$ is logged per cell as a compliance diagnostic and cells below 0.20 are excluded. One forward pass per cell replaces roughly 25 sampled repetitions, because the probability is the quantity the logit likelihood wants. Runs use 4-bit quantization on a free Colab T4.

This also settles a Phase 1–2 nuisance. Temperature 1.0 sampling was treated there as a choice-error process, an assumption the earlier phases could not test. Phase 3 involves no sampling anywhere: the logits are deterministic given the prompt, and all randomness in the design is the randomization the experimenter controls, namely templates, orders, and the payoff grid.

A base and instruct pair differ in weights and in natural elicitation format at the same time, so a parameter gap could be an artifact of format. Every instruct model is therefore run under both formats, its chat template and the identical raw few-shot completion the base model receives. This control was mandatory by design, and it earned its place (Section 5.4).

3.3 The induced-valuation arm

Following the reviewer’s first point (Section 4.3), a second arm asserts a utility function instead of eliciting one. The same 1,680 cells are re-run with an instruction prepended: either risk-neutral expected-value maximization, or expected utility with $u(x) = \sqrt{x}$. Compliance with the induced optimum is scored by `analysis/score_induced.py` as the probability mass placed on the induced-optimal option, reported on cells where the induced rule clearly separates the two options ($|\ln(\text{EU}_g/\text{EU}_s)| \geq 0.05$) and separately on the dominant controls, where both rules agree and no arithmetic is required. This is the instruction-only version of Smith’s (1976) induced-value procedure, without the incentive that makes the original work.

3.4 The reason-then-answer arm

A failure of the induced arm could mean the models cannot compute in an immediate-answer format or that they will not adopt the stated rule. A reason-then-answer arm separates the two. The model generates reasoning greedily, with no sampling and at most 320 tokens, until it writes its own **Answer:** line. The reasoning is kept, **Answer:** is appended, and the A/B distribution is read at that position exactly as in the main arm, so readout and scoring are unchanged. In the chat format the prompt asks the model to think the choice through and end with its answer. In the few-shot format each of the four exemplars gains a one-line reasoning step that states the certain amount or the probability times the payoff for each option, without stating a decision rule. Those

exemplars demonstrate the expected-value arithmetic, so uninduced few-shot reasoning is partly primed toward it, and the cleanest comparisons are within the arm.

Reasoning costs roughly ten times as much compute as an immediate answer, so the arm covers 20 of the 40 gambles per instrument, with every frame, template and order of each (840 cells, fixed seed). All comparisons with the immediate-answer runs below are on these matched cells. Subjects are each model in the format where it passes the dominance control: Qwen Instruct in chat, OLMo SFT and OLMo Instruct in few-shot, each uninduced and under the risk-neutral instruction. The OLMo DPO checkpoint and Llama Instruct are not yet run.

4 Econometric Model and Estimation

4.1 Fractional maximum likelihood

The observed data are probabilities, one per design cell, rather than Bernoulli draws. The Phase 1 and Phase 2 likelihoods carry over unchanged if the Bernoulli log-likelihood is replaced by the binomial deviance with the observed probability as the target, the standard fractional-response objective (Papke and Wooldridge 1996):

$$D(\theta) = \sum_c \left[y_c \ln \frac{y_c}{\hat{p}_c(\theta)} + (1 - y_c) \ln \frac{1 - y_c}{1 - \hat{p}_c(\theta)} \right],$$

where y_c is the elicited probability of the gamble in cell c and $\hat{p}_c(\theta)$ is the structural choice probability. The structural model is the Phase 2 specification: reference-dependent value $v(x) = (x - \rho)^\alpha$ for $x \geq \rho$ and $-\lambda(\rho - x)^\alpha$ below, with $\rho = \phi \cdot \text{anchor}$, Prelec weighting $w(p) = \exp(-(-\ln p)^\gamma)$, a Fechner noise term μ , and the position nuisance parameter δ . The Phase 1 risk cells are the anchor-zero rows of the same likelihood, so one estimator covers both instruments. Fitting is by multi-start Nelder-Mead under the Phase 2 stated-reference restriction $\phi = 1$. Parameter intervals are profile-likelihood intervals from a refit with wide bounds ($\lambda \in [10^{-4}, 8, 103]$, $\gamma \in [0.018, 54.6]$, $\alpha \in (0, 1)$): each of λ , γ and α is fixed on a grid, the remaining parameters are re-optimized, and the interval is read where the scaled profile crosses the χ^2_1 95% value. Because the objective is a quasi-likelihood and the ten presentations of a gamble (five templates, two orders) are not independent, the profile is scaled by a sandwich variance clustered on the 80 gambles. Where a parameter sits at the edge of the wide box the sandwich is invalid, and the scaling falls back to Pearson dispersion, which ignores the clustering and gives intervals that are probably too narrow. A contrast between two subjects combines their two profiles exactly, since they share no parameters. An earlier version of this phase used a bootstrap over design cells; it failed at the old parameter bounds, where one interval overflowed to -3.8×10^{48} .

Validation gate: the estimator recovers the generating parameters near-exactly on the simulated probability agent (`data/phase3_dryrun.csv`), so the degenerate fits reported below are properties of the models' choice patterns rather than estimator failures.

4.2 Model-free quantities and held-out validation

Two quantities need no structural model and are reported as primary throughout. **Dominance compliance** is the mean probability placed on the strictly larger sure amount across the 80 dominant controls, a per-model, per-format engagement check: a subject that fails it is not expressing preferences. **The frame gap** is mean $P(\text{gamble})$ in the mixed frame minus the gain frame over

terminal-wealth-identical lotteries, the Phase 2 headline in probability units. Structural parameters are reported alongside, with boundary flags, for the reason given in Section 8.

Phase 1 closed on an honest weakness: the structural model essentially tied a probability-only logit on in-sample fit and lost on BIC. `analysis/holdout_validation.py` settles it out of sample. The 40 gamble identifiers are split 30/10, each candidate is fit on the training gambles, and mean per-cell held-out deviance is evaluated on cells belonging to gambles the fit never saw, over 10 random splits. Candidates are the structural model, a two-parameter logit in the win probability alone, and the training-mean constant.

4.3 Reviewer feedback and design response

Philip Armour reviewed the Phase 1–2 design memo and returned four points in August 2026. Each is answered by an element of the design above, and the correspondence is worth stating explicitly because three of the four changed what this phase reports.

1. **Induced valuation.** His central point invoked the induced-value tradition: if a stated utility function cannot be induced in the subject, elicited “natural” preferences rest on weaker ground, and if it can, compliance calibrates how far to trust the elicitation. This became the arm in Section 3.4, and it lands harder here than in Phase 1, because the OLMo staircase can ask where in post-training a model becomes the kind of subject induced-value theory presumes. His harder variant, denominating payoffs in a token or computation budget so the stakes are real, is the fall workplan.
2. **The λ - ϕ collinearity.** He pointed to Rees-Jones and Wang (2022) on testing reference points, a minimally parametric approach to the exact degeneracy Phase 2 hit. It is the reading gate for the Phase 2 revision.
3. **Temperature and what is being randomized.** He asked whether the model’s outputs are deterministic given the same prompt. They are, and the Phase 3 readout uses precisely that: no sampling anywhere, with the design randomization (templates, orders, payoff grid) supplying the distribution the bootstrap resamples. His further ask, sampling decision-maker characteristics from CPS or ACS distributions, changes the estimand from the deployed assistant to a simulated population, so it belongs as an additional arm rather than a replacement.
4. **Out-of-sample prediction.** An emphatic yes, and pure analysis on data already in hand. Implemented as Section 4.2; it answers the Phase 1 near-tie (Section 5.8).

The project sits inside the structural behavioral economics tradition DellaVigna (2018) surveys: estimate the parameters of a behavioral model rather than demonstrate a bias, so magnitudes are comparable across designs and the identifying assumptions are stated. That is what makes a stage-level attribution meaningful rather than decorative.

5 Results

5.1 Data inventory and engagement

All eleven uninduced runs completed with a 0% exclusion rate and high A/B token mass (medians 0.908 to 1.000), so no model is failing to treat the prompt as an A/B question. The induced

runs excluded at most 1.8% of cells. Table 1 reports the inventory alongside the two model-free quantities, with the Phase 2 Haiku result as the reference row.

Table 1: Data inventory and model-free quantities. Each run scores the same 1,680-cell grid. A/B mass is the median unnormalized probability on the answer tokens. Dominance is mean probability on the strictly larger sure amount over 80 controls. The frame gap is mixed minus gain uptake on terminal-wealth-identical lotteries.

Subject	Format	A/B mass	Dominance	Gain	Mixed	Frame gap
Qwen2.5-7B base	fewshot	0.960	0.639	0.161	0.365	+0.203
Qwen2.5-7B Instruct	fewshot	0.995	0.680	0.000	0.001	+0.001
Qwen2.5-7B Instruct	chat	1.000	0.927	0.267	0.893	+0.626
OLMo-2-7B base	fewshot	0.908	0.555	0.499	0.598	+0.099
OLMo-2-7B SFT	fewshot	0.982	0.756	0.308	0.551	+0.243
OLMo-2-7B DPO	fewshot	0.986	0.882	0.218	0.536	+0.318
OLMo-2-7B Instruct	fewshot	0.992	0.896	0.159	0.508	+0.349
OLMo-2-7B Instruct	chat	0.985	0.636	0.475	0.552	+0.078
Llama-3.1-8B base	fewshot	0.961	0.623	0.431	0.531	+0.099
Llama-3.1-8B Instruct	fewshot	0.983	0.782	0.444	0.478	+0.034
Llama-3.1-8B Instruct	chat	0.999	0.804	0.589	0.538	−0.051
<i>claude-haiku-4-5</i> (Phase 2)	sampled	—	1.000	0.398	0.096	−0.302

5.2 Base models barely track value

All three base models emit answer tokens with high confidence and then fail the easiest question in the instrument. Dominance compliance is 0.555 for OLMo, 0.623 for Llama, and 0.639 for Qwen, against Haiku’s 1.000 in Phase 2. OLMo base picks the gamble at roughly a coin flip in every frame. The few-shot wrapper is doing what it was built to do, which is to teach the answer format, and it is not producing value-tracking behavior.

This reframes the attribution question the phase was designed to ask. The plan was to test whether post-training bends existing preferences toward or away from human patterns. The base models show there is little to bend: post-training appears to install most of the value-tracking that makes “preferences” a meaningful description of the output at all. A base model at 0.56 dominance compliance does not have unusual preferences, it has no measurable ones, and any structural parameter fit to its choices describes a pattern closer to statistical noise than to taste.

5.3 The staircase: attribution to supervised fine-tuning

The OLMo-2 stages are the core of the phase. All four checkpoints are elicited under the identical few-shot format, which is OLMo’s well-behaved format (Section 5.4), so the format confound is held fixed by construction rather than by argument. Table 2 and Figure 1 report the result.

Base to SFT: everything moves. Dominance compliance rises from 0.555 to 0.756 (+0.200, 95% CI [+0.177, +0.226]). The frame gap more than doubles, from +0.099 to +0.243 (+0.144, CI [+0.133, +0.155]). The structural fit leaves the corner of the parameter space. With the old clip removed, the base model’s λ goes all the way to zero (upper limit 0.03): it puts no weight on the

Table 2: OLMo-2 training staircase, all elicited under the identical few-shot format ($n = 1,680$ cells per stage; structural fit on the 1,600 core cells under $\phi = 1$). Brackets are 95% profile-likelihood intervals from the wide-bounds refit. † = estimate at the edge of the wide box, where λ reaches zero and α reaches one; the base model’s intervals use the unclustered dispersion scaling.

Stage	Dominance	Frame gap	λ	γ	α	δ
Base	0.555	+0.099	0.000 † [0, 0.03]	1.839 [1.65, 2.07]	1.000 † [0.99, 1]	-0.874
SFT	0.756	+0.243	1.142 [1.01, 1.26]	0.378 [0.17, 0.57]	0.061 [0, 0.23]	+1.179
DPO	0.882	+0.318	1.081 [0.96, 1.19]	0.419 [0.24, 0.60]	0.042 [0, 0.20]	+1.073
Instruct	0.896	+0.349	1.057 [0.95, 1.16]	0.349 [0.20, 0.52]	0.035 [0, 0.18]	+1.064

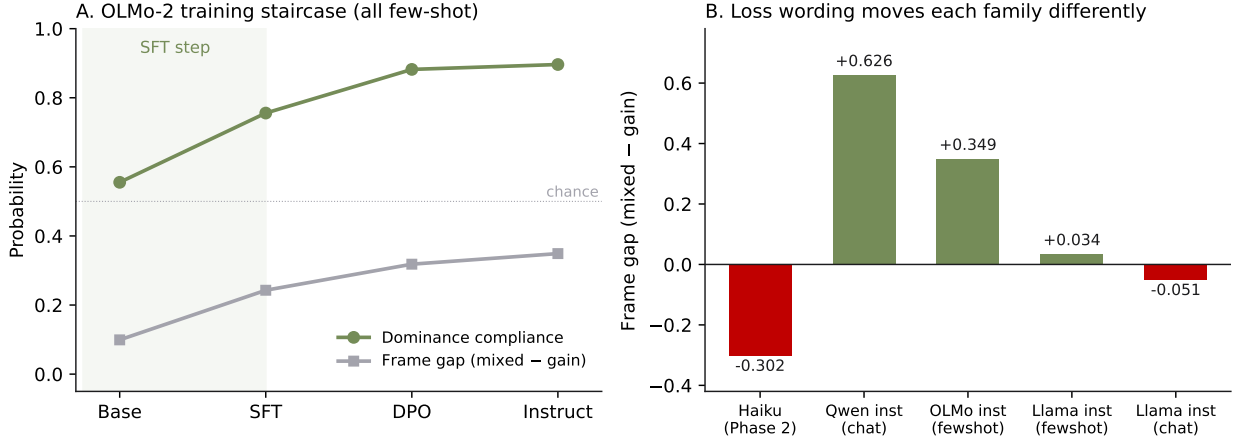


Figure 1: Panel A: the OLMo-2 staircase. Dominance compliance and the frame gap both make their largest move at supervised fine-tuning, and the base model sits near chance on dominance. Panel B: the framing response across families, with the Phase 2 Haiku estimate for comparison. The sign and the magnitude are both family-specific.

loss side of a mixed-frame gamble. SFT moves it to 1.142, $\Delta\lambda = +1.14$ with CI $[+1.02, +1.26]$. Weighting moves from 1.839 to 0.378, $\Delta\gamma = -1.46$ with CI $[-1.79, -1.18]$, and curvature from its upper limit of one to 0.061, $\Delta\alpha = -0.94$ with CI $[-1.00, -0.78]$. The position parameter reverses sign, from -0.874 to $+1.179$. A base model whose fit sits in a corner of the parameter space becomes, after one supervised stage, a subject with an interior fit and a measurable framing response. At SFT and every later stage the α interval runs down to zero, so payoff size barely enters the value function. That is the structural form of the Phase 1 finding that the model prices probability rather than payoffs.

SFT to DPO: parameters hold, behavior keeps moving. Dominance compliance continues to climb, 0.756 to 0.882 (+0.126, CI $[+0.098, +0.152]$), and the frame gap rises to +0.318 (+0.075, CI $[+0.069, +0.082]$). Both changes are clearly nonzero. The preference parameters do not move by an amount the profile-likelihood intervals can detect: $\Delta\lambda = -0.06$ with CI $[-0.23, +0.11]$, $\Delta\gamma = +0.04$ with CI $[-0.24, +0.34]$, $\Delta\alpha = -0.02$ with CI $[-0.22, +0.20]$.

DPO to Instruct: small movement. Dominance 0.882 to 0.896 (+0.014, CI $[+0.003, +0.024]$), frame gap +0.318 to +0.349 (+0.031, CI $[+0.025, +0.036]$). The parameter changes are $\Delta\lambda = -0.02$ with CI $[-0.18, +0.13]$, $\Delta\gamma = -0.07$ with CI $[-0.32, +0.19]$, and $\Delta\alpha = -0.01$ with CI $[-0.20, +0.18]$. Every parameter interval covers zero.

Stage differences in the model-free quantities are paired over identical cells and bootstrapped over the 40 gamble identifiers; because the logit readout is deterministic, these intervals reflect only the design distribution, and they are narrow for that reason. Parameter intervals are the gamble-clustered profile-likelihood intervals of Section 4 (`analysis/profile_refit.py`); Figure 2 shows the profiles themselves.

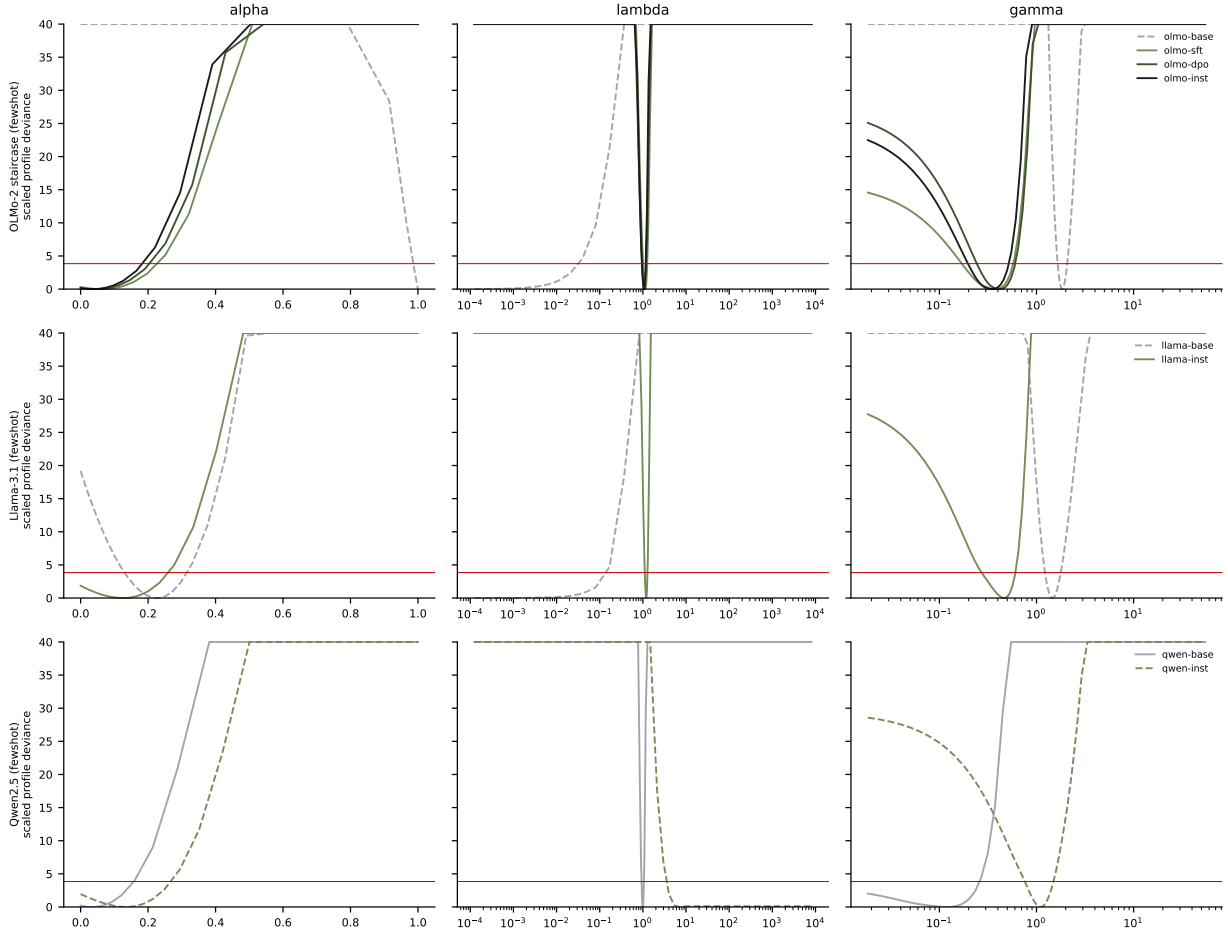


Figure 2: Scaled profile deviance for α , λ and γ across the wide-bounds refit. The horizontal line is the 95% cutoff; a curve that stays under it to the edge of the axis marks a parameter the data do not bound in that direction. Solid curves use the gamble-clustered scaling, dashed curves the dispersion fallback for fits at the edge of the box. Top row: after SFT the three OLMO stages lie on top of one another, while the base model sits apart.

The two sets of results fit together in one way. The structural parameters settle at SFT, while the model-free quantities keep rising through DPO. The later movement shows up as a worse structural fit: deviance on the same 1,600 core cells rises from 229 at SFT to 421 at DPO and 457 at Instruct, and the Pearson dispersion rises with it. Nor is it a fall in choice noise within the model. The Fechner scale μ declines after SFT, but so does α , and with a value function this flat their ratio, which sets how sharply choices respond to value, stays near 0.10 at all three post-trained stages. Preference optimization therefore changes something the specification does not capture, and identifying what it changes is open work.

The central claim is stated in two parts. In this family, on these instruments, SFT is the largest single step and the only one at which the estimated preference parameters move; about three fifths of the base-to-Instruct change in both model-free quantities arrives there. And the measurable economic profile is in place before preference optimization begins. This bears on a literature that has pointed at different stages. Itzhak et al. (2024) found instruction tuning and RLHF amplifying cognitive biases relative to base models; the same group’s later cross-tuning study (Itzhak, Belinkov, and Stanovsky 2025) finds that bias profiles cluster by pretraining backbone, with finetuning bringing them out. The staircase cannot choose between SFT installing the profile and SFT exposing one planted in pretraining, because the base model fails the dominance control and its preferences cannot be measured. What it rules out is that the profile first appears at the preference-optimization stage.

5.4 Format instability, with a model-specific direction

The dual-format arm was included as a control and returned a result. The same Qwen Instruct weights behave as two different subjects under the two formats. In the chat format it reaches 0.927 dominance compliance and a +0.626 frame gap. In the few-shot format it refuses the gamble almost totally (uptake 0.000 in the gain frame, 0.001 in the mixed frame), the frame gap vanishes, the structural λ has no upper limit (the profile is flat from 3.6 out to the edge of the box, the direction the free- ϕ Phase 2 fit also slid), and $\delta = -4.67$, which says the choices are riding position rather than value.

OLMo Instruct shows the same instability in the opposite direction. Few-shot is its well-behaved format (dominance 0.896, frame gap +0.349); the chat format degrades it (dominance 0.636, frame gap +0.078, $\delta = +3.83$). Llama Instruct is the exception that makes the pattern legible: it is roughly stable across both formats, with dominance 0.782 few-shot and 0.804 chat, and frame gaps of +0.034 and -0.051 . That stability holds only for the model-free quantities. The structural parameters move between Llama’s two formats: $\Delta\gamma = +1.84$ with CI $[+1.17, +3.47]$, $\Delta\lambda = +2.14$ with CI $[+0.89, +4.43]$, and α rises from 0.13 to its upper limit of one. OLMo Instruct’s parameters also move across formats ($\Delta\lambda = +0.31$, CI $[+0.20, +0.43]$; $\Delta\gamma = -0.19$, CI $[-0.37, -0.03]$). Figure 3 puts the three families side by side.

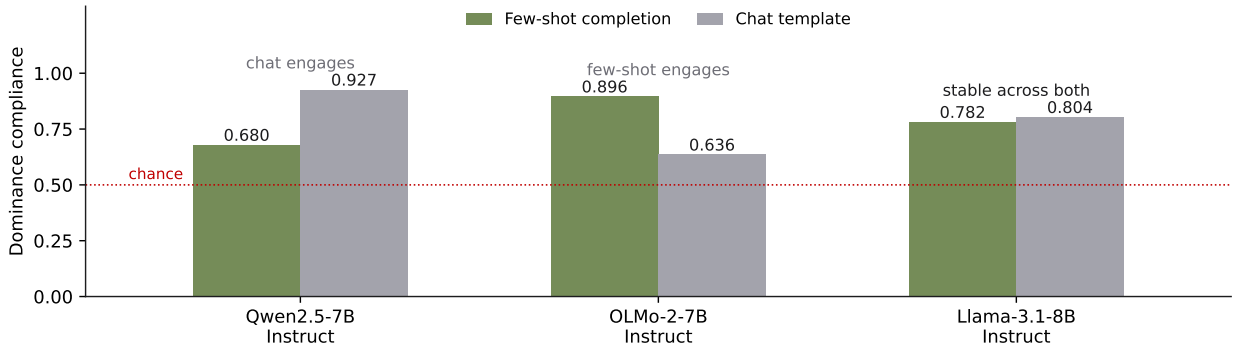


Figure 3: Dominance compliance by elicitation format for the three instruct models. Which format elicits an engaged subject reverses between Qwen and OLMo, and Llama’s dominance compliance is stable across both. The same weights score 0.680 or 0.927 depending only on how the prompt is packaged, so format is part of the measurement rather than a neutral container for it.

Two methodological consequences follow. Format instability generalizes across model families, while which format elicits engaged behavior does not. And a cross-model comparison must hold format fixed *and* verify per model that the fixed format elicits engagement, or it risks comparing an engaged model against one that has disengaged. Dominance compliance is the cheap check, at 80 cells per model per format, but Llama shows it is not a sufficient one: a subject can pass it in both formats and still return different parameters in each. The OLMo staircase in Section 5.3 is all-few-shot, OLMo’s good format, so the staged attribution is unaffected.

The same finding disqualifies one contrast the phase set out to run. The Qwen base-to-instruct attribution under the few-shot format ($\Delta\lambda$ at least +2.7 with no upper limit, $\Delta\gamma$ CI [+0.61, +1.41]) cannot be read as a post-training effect, because the instruct model’s few-shot behavior does not represent that model in its natural format; it is reported as format-confounded and excluded from the attribution claims. The Llama pair, elicited few-shot on both sides of a format-stable instruct model, gives $\Delta\lambda = +1.20$ with CI [+1.04, +1.27] and $\Delta\gamma = -1.00$ with CI [-1.35, -0.72] from base to instruct. As with OLMo, the base model’s λ sits at zero, which is the parameter’s natural floor, so the contrast in levels is well defined; the base side uses the unclustered scaling. Because Llama’s parameters move with format, this contrast holds for the few-shot format and should not be carried to chat.

5.5 Framing heterogeneity across families

Phase 2 found that Haiku suppresses gambling under loss wording, a frame gap of -0.302 , which runs against the textbook prospect-theory direction of risk-seeking in the domain of losses. Every open model here moves the other way or barely moves. Qwen Instruct in chat amplifies strongly, $+0.626$. The OLMo stages amplify moderately, $+0.24$ to $+0.35$. Llama Instruct is nearly frame-invariant, $+0.034$ few-shot and -0.051 chat.

Three families, three responses to the identical manipulation, and a fourth from the frontier model in Phase 2. The cross-model heterogeneity is a stronger result than any single family’s gap. Loss framing is not a fixed property of language models as a class, and the direction of the response is plausibly a function of the post-training recipe. A literature that estimates a prospect-theory parameter on one instruct model and reports it as a fact about LLMs is over-generalizing from a sample of one recipe. The staircase supplies the complementary within-family fact: framing sensitivity is installed early, then grows monotonically but modestly through the later stages.

5.6 Induced valuation: the models cannot execute an induced rule

Table 3 reports compliance with an explicitly instructed decision rule.

Instruction-only induction fails on cells requiring computation, at every training stage, in both families. Compliance mass on expected-value-clear cells never exceeds roughly 0.55 anywhere, close to indifference. The induction effect grows monotonically up the OLMo staircase (from $+0.017$ to $+0.073$ on all core cells, $+0.029$ to $+0.075$ on clear cells) and stays trivial in absolute terms. Instructed dominance compliance moves the opposite way, climbing to 0.950 and 0.969 in probability mass at the DPO and Instruct stages, which is 1.000 in hard accuracy: every dominant control answered correctly. Figure 4 sets the two against each other.

The instruction helps exactly where the induced optimum requires no arithmetic and does nothing where it requires multiplying p by h . The sharpest evidence is the square-root run. Told that $u(x) = \sqrt{x}$, Qwen Instruct’s compliance collapsed to 0.124 hard accuracy on clear cells, far below

Table 3: Induced-valuation compliance under a risk-neutral expected-value instruction. Compliance is the mean probability mass on the induced-optimal option, restricted to cells where the induced rule clearly separates the options ($|\ln \text{EU ratio}| \geq 0.05$; 1,221 to 1,240 cells per run). Baseline is the uninduced run from the same model and format, scored against the same rule.

Subject (format)	EV compliance, clear cells			Dominance compliance	
	Baseline	Induced	Difference	Baseline	Induced
Qwen Instruct (chat)	0.524	0.541	+0.017	0.927	0.886
OLMo base (fewshot)	0.516	0.545	+0.029	0.555	0.593
OLMo SFT (fewshot)	0.493	0.525	+0.032	0.756	0.821
OLMo DPO (fewshot)	0.470	0.533	+0.063	0.882	0.950
OLMo Instruct (fewshot)	0.449	0.524	+0.075	0.896	0.969

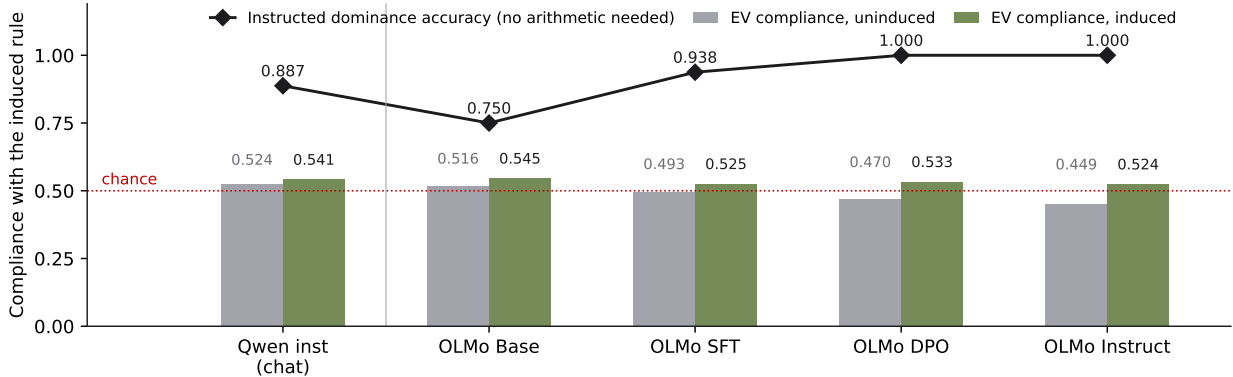


Figure 4: Compliance with an instructed decision rule, by what the rule demands. The bars are expected-value compliance on cells where following the rule requires multiplying p by h : uninduced against induced, both hugging chance at every training stage. The line is accuracy on the dominant controls under the same instruction, where the induced optimum requires no arithmetic, reaching 1.000 by DPO. Post-training buys instruction-following on the cells that need no computation and does not buy the computation.

its own uninduced 0.479, and the errors have a legible pattern: gain-frame accuracy of 0.035 corresponds to near-always gambling, which is what comes of comparing square-root-transformed payoffs while dropping the probability term. The model did something with the instruction. It did not do the instruction.

This supports “cannot compute in this format” over “will not comply.” In an immediate-answer elicitation the model must emit a letter at the next token with no intermediate computation available, while post-training steadily improves its following of the no-computation part of an instruction. The induced-value calibration therefore returns a negative result, and the negative result is informative: it puts a ceiling on how much any immediate-answer elicitation of this kind can claim about deliberate optimization. The decisive test is a reason-then-answer arm that lets the model compute the expected value before committing to a letter. Section 5.7 reports it, and it confirms the inability reading.

5.7 Reasoning first: the ceiling was the format

Table 4 and Figure 5 compare the four conditions on the 840 matched cells.

Table 4: The reason-then-answer arm against the immediate-answer runs on the same 840 cells. EV is the share of the 620 clear core cells on which the model puts more than half its probability on the expected-value-optimal option (hard accuracy, so the immediate-answer figures differ slightly from the compliance mass in Table 3). Gap is mean $P(\text{gamble})$ in the mixed frame minus the gain frame.

Condition	Qwen Instruct (chat)		OLMo SFT (few-shot)		OLMo Instruct (few-shot)	
	EV	Gap	EV	Gap	EV	Gap
Immediate, uninduced	0.532	+0.626	0.519	+0.241	0.497	+0.346
Immediate, induced	0.569	+0.401	0.545	−0.008	0.542	+0.003
Reason, uninduced	0.829	−0.135	0.681	+0.375	0.740	+0.055
Reason, induced	0.965	−0.145	0.713	+0.358	0.803	−0.024

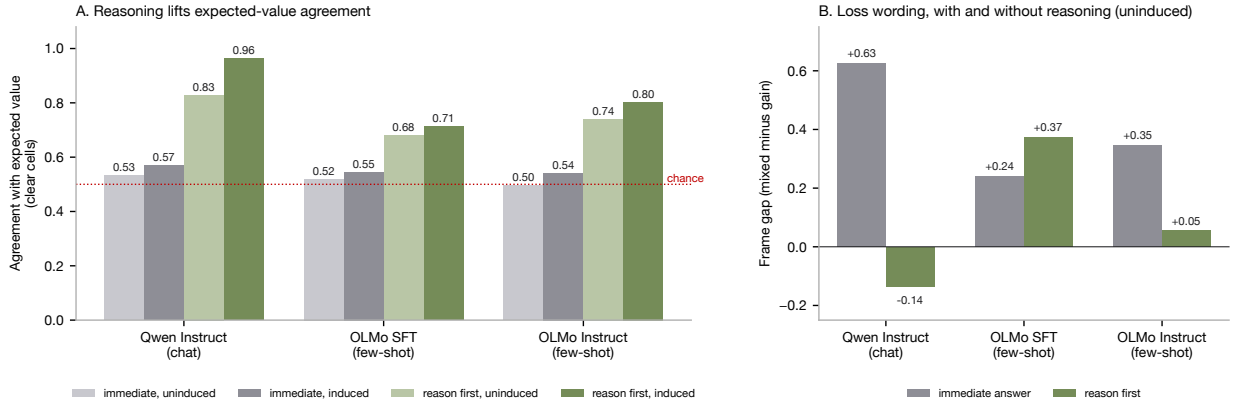


Figure 5: Panel A: agreement with expected value on clear cells, by elicitation regime and instruction. Reasoning moves every subject well above chance, and the instruction adds to it. Panel B: the frame gap without instruction. Reasoning reverses Qwen’s response to loss wording and removes OLMo Instruct’s, while OLMo SFT keeps a large gap.

The models can follow the rule when they can write the arithmetic down. With reasoning allowed, induced expected-value agreement on clear cells rises from 0.569 to 0.965 for Qwen Instruct (+0.395, 95% CI [+0.331, +0.456], bootstrapped over the 40 gamble clusters) and from 0.542 to 0.803 for OLMo Instruct (+0.262, CI [+0.199, +0.325]). Dominance compliance reaches 1.000 for Qwen and 0.975 for OLMo Instruct. The traces show the computation directly: “ $(0.60 \times \$234) + (0.40 \times \$0) = \$140.40 \dots \$140.40 > \$139$.” Reasoning moves the model toward expected value even without the instruction. Every uninstructed Qwen trace frames the choice as an expected-value comparison, and agreement rises from 0.532 to 0.829 (+0.297, CI [+0.213, +0.379]). The instruction-only failure of Section 5.6 was a limit of the immediate-answer format, and the structural parameters of Phases 1 through 3 describe that regime: a fast, intuitive choice process, distinct from the deliberate one the same weights can also run.

Under reasoning, the framing effect becomes an accounting error. Qwen’s frame gap reverses from +0.626 to −0.135 (CI [−0.221, −0.053]), the direction Phase 2 found for Haiku (−0.302).

The mechanism is visible in the traces. In the induced run, 21 of the 22 misses on clear cells fall in the mixed frame, and all 21 are cells where expected value favors the gamble: mixed-frame accuracy is 1.000 when the sure amount is optimal and 0.767 when the gamble is. The model computes the gamble as a change from current wealth and compares it with the sure option as a level, as in “ $E_A = (0.47 \times 227) + (0.53 \times 0) - 98 = 8.69 \dots E_B = 98$,” or it nets the gain and loss correctly and then compares the net change with the sure amount. Either way the reference point shifts between the two options, and the shift always favors the safe choice. OLMo Instruct’s gap falls to about zero (+0.055, CI $[-0.075, +0.183]$). Its mixed-frame misses run in both directions and include plain comparison errors (“Since \$151.84 is less than \$117”) that Qwen does not make. Whether Haiku’s anti-textbook Phase 2 response comes from the same mis-referenced computation is a hypothesis these data raise and do not test.

The staircase depends on the regime. Under immediate answer the structural parameters were set at SFT and the later stages moved nothing detectable (Section 5.3). Under reasoning, the later stages matter. OLMo SFT keeps a large frame gap (+0.375, CI $[+0.286, +0.466]$), larger than its own immediate-answer gap, and by the final Instruct checkpoint it is nearly gone: the difference is -0.320 (CI $[-0.449, -0.201]$). Induced expected-value agreement also rises from SFT to Instruct (0.713 to 0.803, difference +0.090, CI $[+0.030, +0.160]$); the uninduced rise (0.681 to 0.740) is not distinguishable from zero. Preference optimization and the final stage leave the immediate-answer parameters where SFT put them and change how the model reasons. The DPO checkpoint has not been run in this arm, so the change cannot yet be divided between DPO and the final stage.

5.8 Held-out validation

Phase 1 reported honestly that a probability-only logit essentially tied the structural model in sample and beat it on BIC. It does not survive out of sample. Fitting on 30 gambles and predicting cells from the 10 held-out gambles, over 10 random splits, the structural model wins on every split for all three families tested. Mean held-out per-cell deviance is 0.297 for the structural model against 0.401 for the probability-only logit and 0.411 for the constant on Qwen Instruct chat; 0.285 against 0.333 and 0.345 on OLMo Instruct few-shot; and 0.094 against 0.117 and 0.125 on Llama Instruct few-shot. The tally is 10/10 splits in each family (Figure 6), and the ordering is the same on root mean squared error.

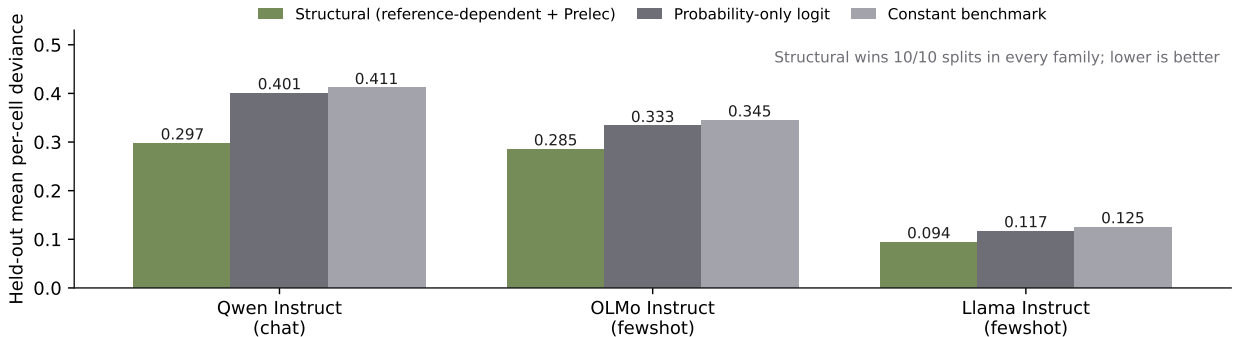


Figure 6: Held-out mean per-cell deviance on gambles the fit never saw, averaged over 10 random 30/10 splits. The structural model beats the probability-only reduced form in every family and on every individual split, and the constant benchmark trails both. Phase 1’s in-sample near-tie between the structural model and the probability-only logit does not survive out of sample.

The structural model therefore earns its place on prediction, not on interpretability alone. Win probability explains a great deal of the behavior, as Phase 1 found. It does not explain the behavior on gambles the fit has never seen.

6 Interpretation

Post-training creates the subject before it shapes the taste. The intuitive picture of a base model with raw preferences that alignment then bends is not what the data show. Base models fail the dominance control at 0.56 to 0.64 while emitting answer tokens confidently, which is a subject producing a plausible letter rather than comparing two amounts of money. Whatever economic behavior is measurable in the instruct models becomes measurable only during post-training. Whether post-training builds it or exposes a disposition laid down in pretraining, the base model’s choices cannot say. Asking where a bias came from presupposes that something was there to be biased.

Supervised fine-tuning comes first, and preference optimization keeps going. Preference-based training is the stage most often blamed for human-like biases, since it is the stage where human judgments enter the weights. In the one family that publishes stage-level checkpoints, the frame gap, the value-tracking, and the interior structural fit are all present after supervised fine-tuning, and the estimated parameters do not move afterward. DPO still matters: it adds roughly a third of the total change in both model-free quantities, through a channel the structural model does not represent. So the ordering is clear, and the division of labor is only partly clear. SFT sets the parameters and does most of the moving; preference optimization amplifies the model-free expression without relocating the estimates. This is one family with one set of instruments, and the base model’s failed dominance control leaves open whether SFT is the origin or only the point of first visibility. The claim also holds only for the immediate-answer regime: when the model reasons first, the later stages remove most of the frame gap SFT leaves (Section 5.7). Either way, the result points research attention at the least glamorous stage, and away from any single stage as the whole story.

The bias is a property of a recipe rather than of a technology. Four systems, four responses to the same loss-wording manipulation, from strong suppression through strong amplification to near indifference. The published work reporting prospect-theory parameters for a single instruction-tuned model, including the closest incumbents (Liu et al. 2025; arXiv:2508.08992), is describing a training recipe. Heterogeneity of this size means an evaluation performed on one model transfers to another only by assumption.

Format is a measurement instrument, and it is not neutral. The same weights produced a coherent economic subject and a near-total gambling refusal depending only on whether the prompt arrived as a chat message or a raw completion, and which format worked reversed between families. Any benchmark comparing models under one fixed format is running an uncontrolled interaction.

The estimand is a regime. In an immediate-answer format the models cannot follow an instructed arithmetic rule; allowed to reason, Qwen follows it on 96.5% of clear cells. Every structural parameter in Phases 1 through 3 is therefore a parameter of an immediate-answer choice process. The same weights also run a deliberate process with different properties: it tracks expected value, it responds to loss wording in the opposite direction for Qwen and hardly at all for OLMo Instruct, and it is shaped by training stages the immediate-answer parameters do not register. Reporting a model’s risk preferences without the elicitation regime omits half of the result. Phase 2’s free- ϕ degeneracy also reappeared here, as Qwen’s few-shot λ running without limit toward the top of its

range, the same direction on the same likelihood surface reached by a different route, which marks it as a property of the specification rather than of any one subject.

7 Policy Relevance

The first audit point is the checkpoint after supervised fine-tuning. Whether SFT installs a behavioral property or first makes it measurable, the checkpoint that follows it is the earliest place the property can be inspected, and in this family it is where most of the change happens. An audit that begins at the preference-optimization stage, which receives most of the governance attention, starts after the parameters are set. Where to correct a property is a separate question this design does not answer: if the disposition is planted in pretraining, changing the supervised data may move its expression without removing it. The method generalizes: any lab retaining intermediate checkpoints can run this diagnostic on any behavioral property a fixed instrument can measure.

Stage-level attribution requires open checkpoints, which is a policy variable. The result exists because one organization published four checkpoints along its training path. No frontier developer does. An evaluation regime that wants to attribute model behavior to training decisions, rather than only document that the behavior exists, depends on staged-checkpoint access for auditors. That access is a policy choice available now, and it is inexpensive relative to what it buys.

Evaluations need engagement checks or they will report randomness as preference. Two results here are ways a plausible-looking evaluation goes wrong. A base model returns confident answer tokens and coherent-looking parameters while failing to prefer \$70 to \$50. An instruct model returns a completely different economic profile under a different prompt format. Both produce publishable-looking numbers, and 80 dominance controls catch both. Any evaluation regime for agentic systems should require the engagement diagnostic alongside the headline, for the same reason a survey reports its response rate.

Do not port a measured bias across models. An agent that suppresses risk-taking under loss wording and one that amplifies it will move in opposite directions on the same procurement menu, the same insurance framing, the same triage prompt. A framing vulnerability documented on one model tells a procurement officer nothing reliable about the model deployed, so the measurement must be repeated per model. That is the argument for cheap standardized instruments of the kind this project builds.

State the elicitation regime. The same model reverses its response to loss wording depending on whether it answers immediately or reasons first. Agentic deployments increasingly let models reason before acting, while most behavioral evaluations still read an immediate answer. An evaluation that does not report which regime it measured cannot be compared with one that does, and a framing vulnerability found in one regime may be absent, or reversed, in the regime that is deployed.

Scope. These are properties of specific open-weight checkpoints under a deterministic logit readout with unincentivized forced choice. They are not claimed as universal traits of language models, and the framing-heterogeneity result is precisely the reason such claims are unsafe.

8 Limitations and Next Phase

Fits at the edge of the parameter space. Five subjects have a parameter at the edge of the wide box: the OLMo and Llama base models and Qwen Instruct in chat put λ at zero, OLMo

base and Llama Instruct in chat put α at one, and Qwen Instruct in few-shot has no upper limit on λ . For these the clustered sandwich is invalid and the intervals use dispersion scaling alone, which ignores the correlation among a gamble’s ten presentations and is probably too narrow. The staircase’s central claims do not rest on these intervals: the post-SFT contrasts are interior and clustered, and the move at SFT is visible in dominance compliance and the frame gap alone. A parametric bootstrap from the fitted model, clustered by gamble, would give the boundary subjects properly calibrated intervals.

The base model cannot be measured, so origin is open. SFT is the first stage at which the profile is measurable, and that is all the staircase can say about its origin. Two follow-ups would separate installation from exposure: a base-model elicitation that passes the dominance control (more exemplars, or dominance items among the few-shot examples), after which the frame gap can be remeasured; and a probe for a framing direction in base-model activations. The second connects to the interpretability leads in docs/.

What DPO changes is unmodeled. Preference optimization moves both model-free quantities while the structural parameters hold and the fit deteriorates. The position parameter barely moves after SFT (+1.18, +1.07, +1.06), so it is not the channel. Candidates include heterogeneity across templates or gambles and a noise process that is not Fechner. Fitting template random effects at each stage is the natural first check.

One instruction phrasing in the induced arm. Each induced rule was tested under a single phrasing. A compliance failure under one phrasing is weaker evidence than a failure under several, and the robustness check is unrun.

The reasoning arm is partial. It covers half the gambles, one instruction phrasing, greedy decoding (one trace per cell), and three subjects; the OLMo DPO checkpoint and Llama Instruct remain to be run, and the DPO run is needed to divide the SFT-to-Instruct change in the reasoning regime between DPO and the final stage. The few-shot reasoning exemplars demonstrate the expected-value arithmetic, which primes the uninduced OLMo runs toward it. The reference-point error is identified from traces read by hand and a rough text check; a systematic coding of which reference point each trace uses, and a test of whether one clarifying line (“compare final dollar amounts”) removes it, would make it a measured mechanism. The token-budget incentive-compatible design remains the step that would complete the induced-value procedure.

Hardware, quantization, and one staircase. All runs use 4-bit quantization on free Colab T4 hardware. The exercise compares checkpoints under identical conditions, so a small uniform quantization effect would not disturb the staircase, though this is unverified against full precision. The stage-level attribution also rests on one family, because OLMo-2 is the only family publishing intermediate checkpoints. A second staged family would turn the central claim into a replication.

Phase 3 has done its job. It carries the Phase 1 and Phase 2 instruments to models whose training history is visible, shows that the measured profile is in place at supervised fine-tuning before preference optimization begins, shows that base models lack the value-tracking that would make the word “preference” apply, documents that the framing response differs in sign across families, and answers the Phase 1 near-tie out of sample. It also found the ceiling of its own elicitation and, with the reasoning arm, raised it: the models can follow a stated rule when they may compute, and their framing response and training-stage profile both depend on whether they do.

References

- [1] DellaVigna, S. (2018). Structural Behavioral Economics. In *Handbook of Behavioral Economics*, Vol. 1, 613–723; NBER Working Paper 24797.
- [2] Holt, C. A., and Laury, S. K. (2002). Risk Aversion and Incentive Effects. *American Economic Review*, 92(5), 1644–1655.
- [3] Itzhak, I., Stanovsky, G., Rosenfeld, N., and Belinkov, Y. (2024). Instructed to Bias: Instruction-Tuned Language Models Exhibit Emergent Cognitive Bias. *Transactions of the ACL*; arXiv:2308.00225.
- [4] Itzhak, I., Belinkov, Y., and Stanovsky, G. (2025). Planted in Pretraining, Swayed by Finetuning: A Case Study on the Origins of Cognitive Biases in LLMs. *Conference on Language Modeling (COLM)*; arXiv:2507.07186.
- [5] Kahneman, D., and Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263–291.
- [6] Köszegi, B., and Rabin, M. (2006). A Model of Reference-Dependent Preferences. *Quarterly Journal of Economics*, 121(4), 1133–1165.
- [7] Liu, J., Yang, Y., and Tam, K. Y. (2025). Evaluating and Aligning Human Economic Risk Preferences in LLMs. *Proceedings of EMNLP 2025*; arXiv:2503.06646.
- [8] CPT Estimation on LLMs (2025). Cumulative Prospect Theory Parameters of Instruction-Tuned Language Models. arXiv:2508.08992.
- [9] Papke, L. E., and Wooldridge, J. M. (1996). Econometric Methods for Fractional Response Variables. *Journal of Applied Econometrics*, 11(6), 619–632.
- [10] Prelec, D. (1998). The Probability Weighting Function. *Econometrica*, 66(3), 497–527.
- [11] Rees-Jones, A., and Wang, K. (2022). An Approach to Testing Reference Points. NBER Working Paper 30773.
- [12] Smith, V. L. (1976). Experimental Economics: Induced Value Theory. *American Economic Review*, 66(2), 274–279.
- [13] Train, K. E. (2009). *Discrete Choice Methods with Simulation* (2nd ed.). Cambridge University Press.
- [14] Tversky, A., and Kahneman, D. (1981). The Framing of Decisions and the Psychology of Choice. *Science*, 211(4481), 453–458.
- [15] Tversky, A., and Kahneman, D. (1992). Advances in Prospect Theory: Cumulative Representation of Uncertainty. *Journal of Risk and Uncertainty*, 5(4), 297–323.
- [16] Wang, Y., Li, X., and Chen, G. (2025). Risk Profiling and Modulation for LLMs. arXiv:2509.23058.
- [17] Wu, G., and Gonzalez, R. (1996). Curvature of the Probability Weighting Function. *Management Science*, 42(12), 1676–1690.